CMSI 3300 – Classwork 2

SOLUTION

Instructions:
This worksheet will not only provide you with practice problems for your upcoming exam, but will concrete your knowledge on the theory of knowledge-based systems and inference.
Specific notes:

- Provide answers to each of the following questions and write your responses in the blanks. If you are expected to show your work in arriving at a particular solution, space will be provided for you.
- Place the names of your group members below:

Group Members:

1. _____
2. _____
3. _____

Problem 1 – Propositional Fundamentals

Time for the warmup! Let's make sure you're comfortable with the fundamentals of propositional logic before moving onto the good stuff…

1.1. Match each concept from Propositional Logic to its intuition / definition. Place the letter of each definition from the bank below next to the concept.

A. A logical variable standing for T/F	B. An instantiation of all propositions to T/F
C. The set of all worlds for some given props.	D. A sentence that is equivalent to True
E. Propositions with some logical connectors	F. For example: $\alpha \wedge \neg\alpha$
G. A sentence with ONLY disjoined props.	H. Logical operator encoding if-then claims
I. When one sentence follows from another	J. A sentence with ONLY conjoined clauses
K. Inference rule that operates on clauses	L. A giant sentence composed of rules + facts

_A_ Proposition
_E_ Logical Sentence
_B_ World
_H_ Implication
_C_ Truth Table
_D_ Valid / Vacuous
_F_ Contradiction
_G_ Clause
_J_ CNF
_I_ Entailment
_K_ Resolution
_L_ Knowledgebase

Recall that the way we arrived at the *resolution* rule for inference is from the formal definition of *entailment* that operates on the definition of *models* $M(\alpha) =$ the set of worlds in which the sentence α is true. Specifically, we say that sentence α *entails* sentence β iff:

$$\alpha \models \beta \Leftrightarrow M(\alpha) \subseteq M(\beta)$$

1.2. In other words, if $\alpha \models \beta$, then knowing α to be true allows us to assume that β is too. To wit, complete the following truth table that will be used in later questions (α is done for you).

World	X	Y	Z	$\boldsymbol{\alpha = X \wedge Z}$	$\boldsymbol{\beta = X \vee Y \vee Z}$	$\boldsymbol{\gamma = \neg Y \vee \neg Z}$
$\boldsymbol{w_0}$	F	F	F	F	F	T
$\boldsymbol{w_1}$	F	F	T	F	T	T
$\boldsymbol{w_2}$	F	T	F	F	T	T
$\boldsymbol{w_3}$	F	T	T	F	T	F
$\boldsymbol{w_4}$	T	F	F	F	T	T
$\boldsymbol{w_5}$	T	F	T	T	T	T
$\boldsymbol{w_6}$	T	T	F	F	T	T
$\boldsymbol{w_7}$	T	T	T	T	T	F

1.3. From the table in 1.2., complete the following questions to learn more about entailment (some are done for you to show you the expected format).

1.3.1. List all worlds $w_i \in M(\alpha)$	$M(a) = \{w_5, w_7\}$
1.3.2. List all worlds $w_i \in M(\beta)$	$M(\beta) = \{w_1, w_2, w_3, w_4, w_5, w_6, w_7\}$
1.3.3. List all worlds $w_i \in M(\gamma)$	$M(\gamma) = \{w_0, w_1, w_2, w_4, w_5, w_6\}$
1.3.4. True or false: $M(\alpha) \subseteq M(\beta)$?	True
1.3.5. True or false: $M(\beta) \subseteq M(\alpha)$?	False
1.3.6. True or false: $\alpha \vDash \beta$?	True
1.3.7. True or false: $\beta \vDash \alpha$?	False

1.4. Notice that sentences β and γ in table 1.2 are *clauses* (disjunctions ∨ of propositions), and suppose we have a CNF knowledgebase composed of them:

$$KB = \beta \wedge \gamma = (X \vee Y \vee Z) \wedge (\neg Y \vee \neg Z)$$

One of the biggest mistakes in using resolution is thinking that *multiple propositions* can cancel out at once when resolving between two clauses. For instance, if we `resolve`(β, γ) in the above, we will **NOT** be able to cancel out the negated Y **AND** Z in order to infer X, i.e., $KB \nvDash X$. To prove this, answer the following questions:

1.4.1. List all worlds $w_i \in M(X)$	$M(X) = \{w_4, w_5, w_6, w_7\}$
1.4.2. List all worlds $w_i \in M(KB)$ (Hint: look at 1.3.2. and 1.3.3.)	$M(X) = \{w_1, w_2, w_4, w_5, w_6\}$
1.4.3. True or false: $M(KB) \subseteq M(X)$?	False
1.4.4. True or false: $KB \vDash X$?	False
1.4.5. What is the resulting clause from the operation `resolve`(β, γ)? (Hint: from your work above, you should find it's NOT X)	The vacuous / valid clause, i.e., the clause that is logically equivalent with True

With the caveat from 1.4. in mind, let's try resolving a few clauses… *if* they can be resolved! Recall that the criteria for resolving two clauses is that they share at least one proposition P, and it is negated in one clause but not the other. The resulting "inferred clause" is the set-union of all remaining propositions between the clauses disjoined together.

1.5. Consider a programmatic implementation of resolution that takes two clauses as input and returns an inferred clause, `resolve(clause1, clause2) => inferred_clause`. For each of the following clause arguments, determine:

1. Whether or not the clauses are *amenable* to resolution, i.e., that it's even possible to resolve the clauses based on the criteria stated above. Place T/F in the "Amenable" column to indicate your answer.
2. If so, determine the inferred clause that would be returned by the resolve function. Place a dash (-) in this column if the clauses were not resolvable. If the resulting clause is vacuous/valid, you may reduce it to the literal "T," and if it's a contradiction, you may write ∅ for the resolved clause.
3. Lastly, determine if your inferred clause was either vacuous/valid (i.e., logically equivalent to True, [$\equiv T$]) or a contradiction [∅]. Again, place a dash (-) in these columns if the clauses weren't resolvable.

#	Clause 1	Clause 2	Amenable?	Inferred Clause	$\equiv T$?	∅?
1.5.1.	$X \vee Y$	$\neg Y \vee Z$	T	$X \vee Z$	F	F
1.5.2.	X	X	F	-	-	-
1.5.3.	$\neg X$	X	T	∅	F	T
1.5.4.	$X \vee Y$	$Y \vee Z$	F	-	-	-
1.5.5.	$X \vee Y \vee Z$	$Z \vee \neg Y$	T	$X \vee Z \vee Z \equiv X \vee Z$	F	F
1.5.6.	$Z \vee \neg Y$	$\neg Z \vee Y$	T	T	T	F
1.5.7.	$X \vee Y$	W	F	-	-	-
1.5.8.	$X \vee Y \vee Z$	$\neg Z \vee \neg Y \vee W$	T	T	T	F

Problem 2 – Guided Inference Exercise

You didn't think you could make it through an entire classwork without me bringing up Forney Industries, did you? Here's a story I've been creating for my upcoming novel that'll help you practice some logic (although it isn't particularly logical itself). Note that although artificial, rules and axioms like those that you'll encode below can be used to create simple Q&A systems in practice, some of which you'll use on your first homework.

> Preceding the Great Forneybot Uprising of 2024, the world was in a utopic state of bliss with humankind finally served by throngs of our robotic servants. Forney Industries, sadly, had a number of fatal oversights in their robots' design, any number of whose small, environmental quirks could cause them to violently malfunction. Before the flaws were discovered, it was, of course, too late -- however, it has long been a matter of historical discourse for what event actually triggered the first mass malfunction...

The following story is written from the future perspective, with bracketed propositions [P] indicating the atoms you'll use in the sentences that follow.

- Rules generally follow an "if-then" pattern of implication $if \Rightarrow then$ but statement of multiple facts are usually conjunction $fact_1 \wedge fact_2 \wedge \ldots$
- "If and only if" statements, or variables that follow one another hand-in-hand tend to follow the pattern of biconditionals $\alpha \Leftrightarrow \beta$

2.1. For each of the following plain-English descriptions of a knowledgebase's rules and axioms, provide the propositional logic sentence translation that uses the bracketed propositions [P]. Don't worry about converting these into CNF clauses just yet, you'll do that in the next step!

2.1.1. "Forneybots were found to malfunction $[M]$ *if and only if* they suffer water damage $[D]$ or overheard a logical paradox $[P]$. This one is done for you as an example."

$$M \Leftrightarrow D \vee P$$

2.1.2. "Forneybots may have been caught in a sudden rainstorm $[R]$, which, *as a rule*, only happen during Winter $[W]$ or Summer$[S]$, though *not all* Winters and Summers had rain."

$$R \Rightarrow (W \vee S)$$

2.1.3. "It goes without saying that rainstorms $[R]$ and water damage $[D]$ would have gone hand in hand (i.e., are found together), since the Forneybots were not aware of their water weakness."

$$R \Leftrightarrow D$$

2.1.4. "Logical paradoxes $[P]$, on the other hand, would have only been heard during election season (through TV or radio broadcast), which are held in the Fall $[F]$ (and so all Fall seasons thus came hand in hand with paradoxes—zing!).

$$P \Leftrightarrow F$$

2.1.5. "It also goes without saying that *<u>if</u>* it is one season, *<u>then</u>* it cannot simultaneously be the others." (Hint: there are 3 seasons in this scenario, so you'll need the rule to be repeated 3 times and then logically conjoined!).

$$\big(F \Rightarrow (\neg S \wedge \neg W)\big) \wedge \big(S \Rightarrow (\neg F \wedge \neg W)\big) \wedge \big(W \Rightarrow (\neg S \wedge \neg F)\big)$$

2.1.6. "The Great Forneybot Uprising occurred from a mass malfunction on Nov. 16, 2024." (Hints: there are *multiple* facts being stated here, and they are *axiomatic,* meaning it is *not* a *rule* being stated).

$$M \wedge F$$

Now that we've got our sentences in propositional logic format, we're one step from using them to answer questions about our scenario: first, they need to be converted into Conjunctive Normal Form (CNF), meaning that our KB is just one giant sentence consisting of a *conjunction of clauses.*

2.2. For each of the sentences you found in 2.1., we'll now convert them into CNF and *enumerate* the resulting *clauses* as KB1, KB2, … with the assumption that our final CNF KB will be of the format: $KB = KB1 \wedge KB2 \wedge KB3 \wedge \ldots$ The correct number of resulting clauses from each conversion is given in the answer box. *<u>Show your work in performing each conversion in the space provided.</u>* The first conversion, and expected format of answers, is done for you. Hint: you might use *logical equivalences* to reduce the number of resulting clauses for some conversions.

<table>
<tr><th>Sentence from 2.1.</th><th colspan="3">CNF Conversion</th></tr>
<tr><td rowspan="2">2.1.1.

$M \Leftrightarrow D \vee P$</td><td colspan="3">$M \Leftrightarrow D \vee P$
$\equiv \big(M \Rightarrow (D \vee P)\big) \wedge \big((D \vee P) \Rightarrow M\big)$
$\equiv \big(\neg M \vee (D \vee P)\big) \wedge (\neg(D \vee P) \vee M)$
$\equiv (\neg M \vee D \vee P) \wedge \big((\neg D \wedge \neg P) \vee M\big)$
$\equiv (\neg M \vee D \vee P) \wedge (\neg D \vee M) \wedge (\neg P \vee M)$</td></tr>
<tr><td>$KB1 =$
$\neg M \vee D \vee P$</td><td>$KB2 =$
$\neg D \vee M$</td><td>$KB3 =$
$\neg P \vee M$</td></tr>
</table>

<table>
<tr><td>2.1.2.

$R \Rightarrow (W \vee S)$</td><td colspan="3">$R \Rightarrow (W \vee S) \equiv \neg R \vee (W \vee S)$
$\equiv \neg R \vee W \vee S$</td></tr>
<tr><td></td><td colspan="3">$KB4 = \neg R \vee W \vee S$</td></tr>
<tr><td>2.1.3.

$R \Leftrightarrow D$</td><td colspan="3">$R \Leftrightarrow D \equiv (R \Rightarrow D) \wedge (D \Rightarrow R)$
$\equiv (\neg R \vee D) \wedge (\neg D \vee R)$</td></tr>
<tr><td></td><td colspan="2">$KB5 = \neg R \vee D$</td><td>$KB6 = \neg D \vee R$</td></tr>
<tr><td>2.1.4.

$P \Leftrightarrow F$</td><td colspan="3">$P \Leftrightarrow F \equiv (P \Rightarrow F) \wedge (F \Rightarrow P)$
$\equiv (\neg P \vee F) \wedge (\neg F \vee P)$</td></tr>
<tr><td></td><td colspan="2">$KB7 = \neg P \vee F$</td><td>$KB8 = \neg F \vee P$</td></tr>
<tr><td>2.1.5.

$(F \Rightarrow (\neg S \wedge \neg W))$
$\wedge (S \Rightarrow (\neg F \wedge \neg W))$
$\wedge (W \Rightarrow (\neg S \wedge \neg F))$</td><td colspan="3">$(F \Rightarrow (\neg S \wedge \neg W)) \wedge (S \Rightarrow (\neg F \wedge \neg W)) \wedge (W \Rightarrow (\neg S \wedge \neg F))$
$\equiv (\neg F \vee (\neg S \wedge \neg W)) \wedge (\neg S \vee (\neg F \wedge \neg W)) \wedge (\neg W \vee (\neg S \wedge \neg F))$
$\equiv (\neg F \vee \neg S) \wedge (\neg F \vee \neg W) \wedge (\neg S \vee \neg F) \wedge (\neg S \vee \neg W)$
$\wedge (\neg W \vee \neg S) \wedge (\neg W \vee \neg F)$

[!] Note: because of associativity, there are only 3 unique clauses from the above, without loss of generality:
$\equiv (\neg F \vee \neg S) \wedge (\neg F \vee \neg W) \wedge (\neg S \vee \neg W)$</td></tr>
<tr><td></td><td>$KB9 = \neg F \vee \neg S$</td><td>$KB10 = \neg F \vee \neg W$</td><td>$KB11 = \neg S \vee \neg W$</td></tr>
<tr><td>2.1.6.

$M \wedge F$</td><td colspan="3">[!] This is already in CNF! There are simply two atomic clauses, one for each of the propositions.</td></tr>
<tr><td></td><td colspan="2">$KB12 = M$</td><td>$KB13 = F$</td></tr>
</table>

2.3. Now that we have all of our KB's starting rules and axioms, it's time for inference! In the following problem, we'll use proof by contradiction to prove that it was a logical paradox $[P]$ and NOT water damage $[D]$ that caused the Great Forneybot Uprising. Specifically, we will show:

$$\alpha = \neg D \wedge P$$

Complete the following steps in the next table to prove that $KB \vDash \alpha$:

1. Collect all of your answers from 2.2. in the "Initial Clauses" column, clauses KB1-13
2. Add the negation of the query, $\neg\alpha$, as clause KB14 (Hint: remember DeMorgan)
3. In the "Inferred Clauses" column, perform a search for the empty clause (Ø = contradiction) by resolving pairs of clauses that are amenable to resolution. You may stop once you've found a contradiction, but must indicate which pairs were used to obtain any clauses KB15+ (e.g., if you resolved clauses KB1 and KB2, you would write [KB1, KB2] in that row of the "Resolvents" column, and then the clause they resolve to in the same row of the "Inferred Clauses" column). *You may not need to use all slots for KB15-28, but won't need more than KB28 to solve the problem.* Suggestion: solve the problem on scratch paper before translating your final answer here.

Initial Clauses		**Resolvents**	**Inferred Clauses**
$KB1 = \neg M \vee D \vee P$		KB9, KB13	$KB15 = \neg S$
$KB2 = \neg D \vee M$		KB10, KB13	$KB16 = \neg W$
$KB3 = \neg P \vee M$		KB4, KB15	$KB17 = \neg R \vee W$
$KB4 = \neg R \vee W \vee S$		KB16, KB17	$KB18 = \neg R$
$KB5 = \neg R \vee D$		KB6, KB18	$KB19 = \neg D$
$KB6 = \neg D \vee R$		KB8, KB13	$KB20 = P$
$KB7 = \neg P \vee F$		KB14, KB19	$KB21 = \neg P$
$KB8 = \neg F \vee P$		KB20, KB21	$KB22 = \emptyset$ [!] Contradiction!
$KB9 = \neg F \vee \neg S$			$KB23 = unused$
$KB10 = \neg F \vee \neg W$			$KB24 = unused$
$KB11 = \neg S \vee \neg W$			$KB25 = unused$
$KB12 = M$			$KB26 = unused$
$KB13 = F$			$KB27 = unused$
$KB14 = D \vee \neg P$			$KB28 = unused$

[!] The above represents one POSSIBLE derivation of the empty clause; there are others.

CMSI 3300 – Classwork 3

SOLUTION

Instructions:
This worksheet will not only provide you with practice problems for your upcoming exam, but will add to your toolset as initiate data scientists taking various probabilistic inference problems from start to finish. Specific notes:

- Provide answers to each of the following questions and write your responses in the blanks. If you are expected to show your work in arriving at a particular solution, space will be provided for you.
- Place the names of your group members below:

Group Members:

1. _____
2. _____
3. _____

Problem 1 – The Bayesics

Before we get crazy, let's make sure you're able to use the fundamental rules of probability calculus—these simple rules will appear time and again throughout your AI track!

1.1. Given that $P(A = 1|B = 1) = 0.4$ and $P(B = 1) = 0.2$, find $P(A = 1, B = 1)$ using the *conditioning* rule. Show your work and all symbolic steps before plugging in numbers.

$$P(A = 1, B = 1) = P(A = 1|B = 1)P(B = 1) = 0.4 * 0.2 = 0.08$$

1.2. Given the following joint over C, D, find $P(C = 1)$ using the *law of total probability (LoTP).* Show your work and all symbolic steps before plugging in numbers.

$P(C, D)$	$D = 0$	$D = 1$
$C = 0$	0.3	0.2
$C = 1$	0.4	0.1

$$P(C = 1) = \sum_d P(C = 1, D = d)$$
$$= P(C = 1, D = 0) + P(C = 1, D = 1) = 0.4 + 0.1 = 0.5$$

1.3. Using the table in 1.2., prove, using the definition of independence, that $C \not\perp\!\!\!\perp D$ (i.e, that they are *dependent*) [Hint: you only need to show that independence doesn't hold for one case]

$$C \perp\!\!\!\perp D \iff P(C, D) = P(C) * P(D) \; \forall \, c, d \in C, D$$
$$P(C = 0, D = 0) = 0.3$$
$$P(C = 0) * P(D = 0) = 0.5 * 0.7 = 0.35 \neq P(C = 0, D = 0)$$
$$\therefore C \not\perp\!\!\!\perp D$$

Suppose that a system with 3 binary variables, X, Y, Z has the property that $X \perp\!\!\!\perp Z \mid Y$.

$P(X, Y, Z)$	$X = 0$		$X = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$Z = 0$	0.10	0.05	0.15	0.20
$Z = 1$	0.10	0.05	0.15	0.20

1.4. Showing each step, solve for $P(X = 0|Y = 0, Z = 0)$ and $P(X = 0|Y = 0)$ separately. You may need to use both conditioning and law of total probability.

$$P(X = 0|Y = 0, Z = 0)$$
$$= \frac{P(X = 0, Y = 0, Z = 0)}{P(Y = 0, Z = 0)}$$
$$= \frac{0.1}{0.25}$$
$$= 0.4$$

$$P(X = 0|Y = 0)$$
$$= \frac{P(X = 0, Y = 0)}{P(Y = 0)}$$
$$= \frac{0.2}{0.5}$$
$$= 0.4$$

Problem 2 – Probabilistic Reasoning

Forney Industries is branching further into the medical domain and has made the discovery of a new, rare disease, Schistoforneymiosis. It is believed that roughly 1/1000 of the population suffers from the disease [D], experiencing a consistently wet left foot and sentient freckles.

To test for the condition, the test [T] is an elaborate procedure involving multiple probes, and returns an end result that is either positive (T = True) or negative (T = False). The problem is that the tests are not perfect, but 95% of people who have the disease will test positive, and 2% of people who do *not* have the disease will test positive.

Answer the following questions to answer an important query: the probability that someone has the disease If they test positive. (NB: 80% of MDs fail to answer this question correctly)

2.1. Sometimes, the query that we are interested in answering is of the format P(A|B), but the information we have is in the format P(B|A). Towards this end, we can use *Bayes' Theorem,* (the namesake of Bayesian Networks) which states:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Below, prove Bayes' Theorem using the conditioning rule from probabilistic logic and the fact that $P(A,B) = P(B,A)$.

$$P(A,B) = P(B,A)$$
$$P(A|B)P(B) = P(B|A)P(A)$$
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

2.2. Now that we have Bayes' Theorem under our belt, we'll encode the claims of probabilities from the problem statement into the syntax of probabilistic reasoning. Use binary variables $D = \{0,1\}$ for whether or not someone has the disease, and $T = \{0,1\}$ for whether or not someone tests positive.

1. "It is believed that roughly 1/1000 of the population suffers from the disease"

   P(D = 1) = 0.001

2. “95% of people who have the disease will test positive”

   $P(T = 1 \mid D = 1) = 0.95$

3. “2% of people who do *not* have the disease will test positive”

   $P(T = 1 \mid D = 0) = 0.02$

4. Compute, using the law of total probability, the chance that an arbitrary individual will have tested positive for Schistoforneymiosis.

$$
\begin{aligned}
P(T = 1) &= \sum_d P(T = 1, D = d) \\
&= \sum_d P(T = 1 | D = d) P(D = d) \\
&= P(T = 1 | D = 0) P(D = 0) + P(T = 1 | D = 1) P(D = 1) \\
&= 0.02 * 0.999 + 0.95 * 0.001
\end{aligned}
$$

$P(T = 1) \approx 0.021$

We now have all of the parts needed to solve our problem.

2.3. Compute, using your work from parts 2.1 and 2.2, the probability that a patient has the disease given that they tested positive for it.

$$
\begin{aligned}
P(D = 1 | T = 1) &= \frac{P(T = 1 | D = 1) P(D = 1)}{P(T = 1)} \\
&= \frac{0.95 * 0.001}{0.021}
\end{aligned}
$$

$P(D = 1 \mid T = 1) \approx 0.045$

Note: that’s pretty rare given what is quite the accurate test! The importance of knowing probability calculus!

Problem 3 – Simple Bayesian Network Inference

Forney Industries are committed to the treatment of Schistoforneymiosis. Towards this end, their other genetic mapping division, FornGene, has attempted to isolate certain genetic expressions that contribute to whether or not an individual has a hereditary disposition to the disease. They conduct a study measuring whether or not an individual's female parent [F] and male parent [M] have the genetic marker, and subsequently, if their child [C] has the disease.

In this section, we will attempt to craft a Bayesian Network that explains the data collected by the FornGene group, and use that to answer inference queries related to the disease.

3.1: In the following table, we track each combination of M, F, and C witnessed from the group's study. From this data, construct the joint distribution over $P(M, F, C)$ in the table below. [This gives you some fundamental experience with data science!]

M	F	C	# of data points	$\boldsymbol{P(M, F, C)}$
0	0	0	864	0.432
0	0	1	96	0.048
0	1	0	72	0.036
0	1	1	168	0.084
1	0	0	448	0.224
1	0	1	192	0.096
1	1	0	32	0.016
1	1	1	128	0.064

3.2: We now want a model to interpret this data. In particular, we have a pretty good idea of the nature of cause and effect implicit in this environment (from what we know of the science behind hereditary genetics). Encode these assumptions as independence statements:

1. "The genetic expression of one parent is unaffected by the other"

   $M \perp\!\!\!\perp F$

2. "Whether or not a child has the disease provides information between the parents"

   $M \not\perp\!\!\!\perp F \mid C$

3.3: With the independence claims you're making above, draw the *structure* of a Bayesian Network that encodes these claims.

$$M \rightarrow C \leftarrow F$$

3.4: With the independence claims and structure from Parts 3.2 & 3.3, encode the Bayesian Network's *semantics,* i.e., the Conditional Probability Tables (CPTs), for each variable in the network. [Hint: you will only need the joint distribution, conditioning, and law of total probability to complete each of the following]

1. CPT for *M*

Calculations

$$P(M) = \sum_{f} \sum_{c} P(M, F, C)$$

CPT

M	P(M)
0	0.6
1	0.4

2. CPT for *F*

Calculations

$$P(F) = \sum_{m} \sum_{c} P(M, F, C)$$

CPT

F	P(F)
0	0.8
1	0.2

3. CPT for *C*

Calculations

$$P(F,M) = \sum_c P(M,F,C)$$

$$P(C|F,M) = \frac{P(M,F,C)}{P(M,F)} = \frac{P(M,F,C)}{\sum_c P(M,F,C=c)} \quad \text{OR} \quad = \frac{P(M,F,C)}{P(M)P(F)}$$

CPT

M	F	C	P(C|M, F)
0	0	0	0.9
0	0	1	0.1
0	1	0	0.3
0	1	1	0.7
1	0	0	0.7
1	0	1	0.3
1	1	0	0.2
1	1	1	0.8

3.5: For each of the following inference queries, determine which are *immediately answerable* (or can be reduced, using *d-separation* to those that are) from the previous two sections, and for those that are, provide their value.

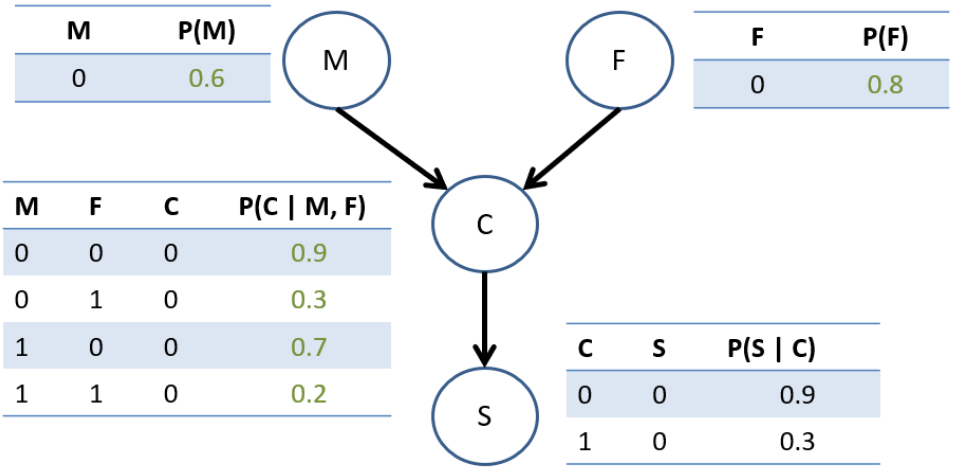
Query	Immediately Answerable or Reducable to such?	If so, value:
$P(M = 0)$	Immediate	0.6
$P(F = 1 \mid M = 1)$	Reducable to $P(F = 1)$	0.2
$P(C = 1)$	Neither	-
$P(C = 0 \mid M = 1)$	Neither	-
$P(C = 0 \mid M = 1, F = 1)$	Immediate	0.2
$P(C = 1 \mid M = 1, F = 1)$	Immediate	0.8

3.6: Given the above, suppose a patient comes in who we do *not* know whether or not they have the disease, but we know that both parents possess the predictive genetic trait (i.e., $M = 1, F = 1$). Should we give this patient the (expensive) drug for Schistoforneymiosis?

Yes, give the drug since $P(C = 0 \mid M = 1, F = 1) < P(C = 1 \mid M = 1, F = 1)$ by quite a bit.

Problem 4 – Exact Bayesian Network Inference

Suppose we now add the presence of a child's symptomology to the network established in the previous section (you didn't read ahead did you?). Note we can model this as an effect of the disease's presence in the child, as follows.



M	P(M)
0	0.6

F	P(F)
0	0.8

M	F	C	P(C | M, F)
0	0	0	0.9
0	1	0	0.3
1	0	0	0.7
1	1	0	0.2

C	S	P(S | C)
0	0	0.9
1	0	0.3

1. Fill in the missing CPT likelihoods from your work in Problem 3.
2. Using these, compute the likelihood that a patient's father had the disease, given that the mother did and the child exhibits symptoms, i.e., find $P(F = 1|M = 1, S = 1)$

Step 1: Label Vars:

$$Q = \{F\}, e = \{M = 1, S = 1\}, Y = \{C\}$$

Step 2: Compute $\boldsymbol{P(Q, e) = P(F, M = 1, S = 1)}$

$$P(F, M = 1, S = 1) = \sum_c P(F, M = 1, S = 1, C = c)$$

$$= P(M = 1)P(F) \sum_c P(C = c|M = 1, F)\, P(S = 1|C = c)$$

- **For F = 0:** $P(F = 0, M = 1, S = 1) = 0.4 * 0.8 * [0.7 * 0.1 + 0.3 * 0.7] = 0.0896$
- **For F = 1:** $P(F = 1, M = 1, S = 1) = 0.4 * 0.2 * [0.2 * 0.1 + 0.8 * 0.7] = 0.0464$

Step 3: Compute $\boldsymbol{P(e) = P(S = 1, M = 1)}$

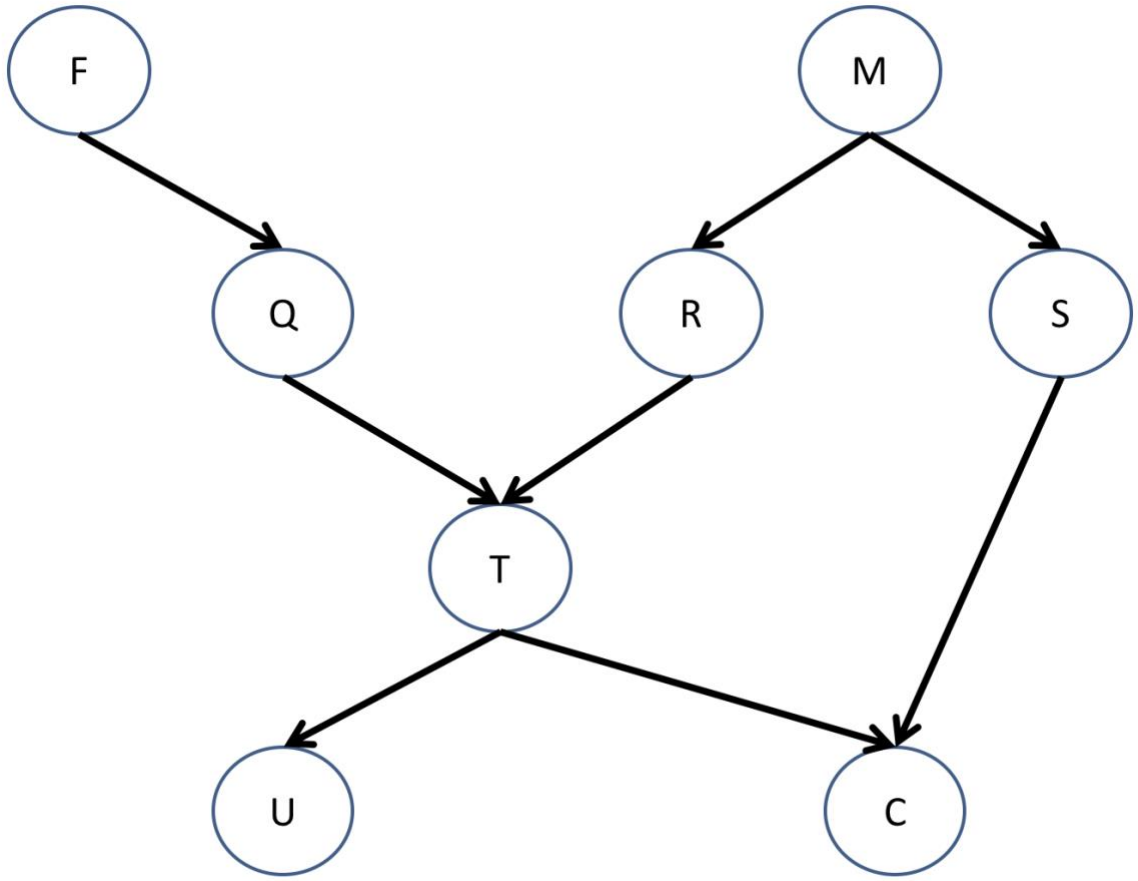
$$P(S = 1, M = 1) = \sum_f P(F = f, S = 1, M = 1) \; = 0.0896 + 0.0464 = 0.136$$

Step 4: Normalize to solve: $\boldsymbol{P(Q|e) = P(Q, e)/P(e)}$

$$P(F = 1|M = 1, S = 1) = \frac{P(F = 1, M = 1, S = 1)}{P(M = 1, S = 1)} = \frac{0.0464}{0.136} \approx 0.341$$

Problem 5 – d-separation

The FornGene group further investigated the genetic relationship between parent and child with regards to the disease's expression and revealed a more complex network than originally thought. In particular, they discovered that a variety of other environmental factors would contribute to an arbitrary patient's risk of attaining the disease, as given by the structure below:



Determine if the following independence relationships hold in the network above:

X	**Y**	**{Z}**	**X ⫫ Y | {Z}?**
F	T	{Q}	True / Yes
Q	R	{T}	False / No
F	M	{C}	False / No
U	S	{C, M}	False / No
Q	C	{T}	False / No
Q	C	{T, S}	True / Yes

CMSI 3300 – Classwork 4

SOLUTION

Instructions:
This worksheet will not only provide you with practice problems for your upcoming exam, but will add to your deep understanding of the mechanics of many probabilistic reasoning systems, managing priorities through utility theory, and curiosity through VPI.

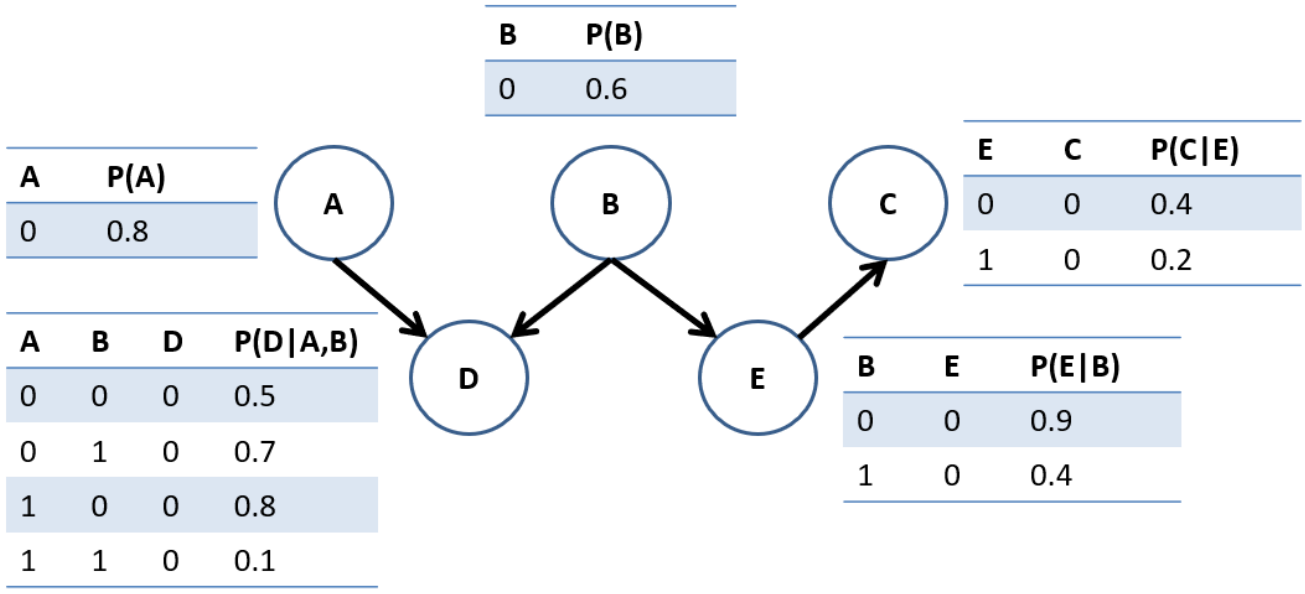
- Provide answers to each of the following questions and write your responses in the blanks. If you are expected to show your work in arriving at a particular solution, space will be provided for you.
- Place the names of your group members below:

Group Members:

1. _____
2. _____
3. _____

Problem 1 – Bayesian Network Exact Inference

Consider the following Bayesian Network and use it to answer the questions that follow.



B	P(B)
0	0.6

A	P(A)
0	0.8

E	C	P(C|E)
0	0	0.4
1	0	0.2

A	B	D	P(D|A,B)
0	0	0	0.5
0	1	0	0.7
1	0	0	0.8
1	1	0	0.1

B	E	P(E|B)
0	0	0.9
1	0	0.4

While examining Exact Inference in Bayesian Networks, we saw some methods for *simplifying* queries and the resulting computations that can yield large performance improvements when implemented. E.g., variables whose CPTs never affect the query outcome can be ignored.

That rule: *To computer $P(Q|e)$, you require ONLY the CPTs of variables that are NOT ancestors of EITHER the Query Q NOR the evidence e. i.e., any hidden variable not meeting this condition are marginalized during step 2 of enumeration inference.*

For example, to answer the query $P(A|B = b, D = d)$, we do not need the CPTs for C nor E because they are not ancestors of A, B, or D in the network. A brief proof using the normalizing constant $\alpha = \frac{1}{P(e)}$:

CPTs Used: ✓ A ✓ B C ✓ D E

Justification:

$$P(A|b,d) = \alpha \sum_e \sum_c P(A,b,c,d,e)$$
$$= \alpha \sum_e \sum_c P(A)P(d|A,b)P(b)P(e|b)P(c|e)$$
$$= \alpha\, P(A)P(d|A,b)P(b) \sum_e P(e|b) \sum_c P(c|e)$$
$$= \alpha P(A)P(d|A,b)P(b)$$

[!] Note how the above is more computationally desirable – no loops / sums required!

1. Using the Bayesian Network and enumeration inference simplification on the previous page, find the solution to the query:

$$P(A = 0|B = 1, D = 1)$$

1.1. Label variables using the notation Q = queries, e = evidence, and Y = hidden vars.

Q = { A }	e = { B=1, D=1 }	Y = { C, E }

1.2. Find $P(Q, e) = \sum_y P(Q, e, y)$, substituting into this formula the variable labels in 1.1.

$$P(A|B = 1, D = 1) = \alpha \sum_e \sum_c P(A, B = 1, D = 1, C = c, E = e) = \alpha P(A)P(d|A, b)P(b)$$

$$A = 0 \Rightarrow P(A, b, d) = P(A = 0)P(D = 1|A = 0, B = 1)P(B = 1) = 0.8 * 0.3 * 0.4 = 0.096$$

$$A = 1 \Rightarrow P(A, b, d) = P(A = 1)P(D = 1|A = 1, B = 1)P(B = 1) = 0.2 * 0.9 * 0.4 = 0.072$$

1.3. Find $P(e) = \sum_q P(Q, e)$.

$$\begin{aligned} P(e) &= P(B = 1, D = 1) \\ &= \sum_a P(a, b, d) = P(A = 0, B = 1, D = 1) + P(A = 1, B = 1, D = 1) \\ &= 0.096 + 0.072 = 0.168 \end{aligned}$$

1.4. Solve the original query, $P(Q|e) = P(Q, e)/P(e)$

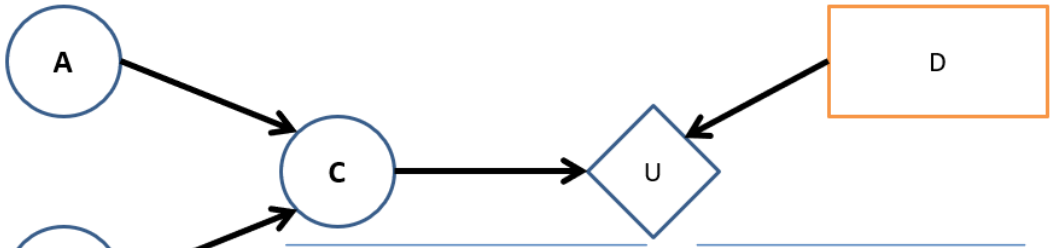
$$P(A = 0|B = 1, D = 1) = \frac{P(A = 0, B = 1, D = 1)}{P(B = 1, D = 1)} = \frac{0.096}{0.168} \approx 0.571$$

Problem 2 – Decision Networks & MEU

Use the following Decision Network with chance nodes A, B, C, decision node $D \in \{0,1\}$, and utility node U to answer the questions that follow.

A	P(A)
0	0.2

B	P(B)
0	0.5



A	B	C	P(C|A,B)
0	0	0	0.8
0	1	0	0.6
1	0	0	0.1
1	1	0	0.5

C	D	U(C, D)
0	0	100
0	1	20
1	0	40
1	1	80

2. Find the $MEU(B = 0)$ (i.e., the Maximum Expected Utility with evidence $B = 0$). The individual steps for computing this are outlined for you below.

2.1. Begin by finding $P(S = s|do(D = d), e)$ $\forall\, s \in S, d \in D$ using Enumeration Inference for decision nets.

2.1.1. (lol nested problem depth) Label your variables for $P(S = s|do(D = d), e)$

Q = { C }	e = { B = 0 }	D = { D }	Y = { A }

2.1.2. Find $P\big(Q, e|do(D)\big) = \sum_y P\big(Q, e, y|do(D)\big)$ (just the formula, we'll plug n' chug next)

$$P\big(C, B = 0|do(D)\big) = \sum_a P(C, B = 0, A = a)$$
$$= \sum_a P(A = a)P(B = 0)P(C|B = 0, A = a)$$
$$= P(B = 0)\sum_a P(A = a)P(C|B = 0, A = a)$$

2.1.3. Using your formula from the previous problem (which should have found that the chance nodes were independent of the decision), find $P(Q, e|do(D))\ \forall\ q \in Q$

$$C = 0 \Rightarrow\ P(C = 0, B = 0|do(D)) = P(B = 0)\sum_a P(A = a)P(C = 0|B = 0, A = a)$$
$$= 0.5[0.2 * 0.8 + 0.8 * 0.1]$$
$$= 0.12$$
$$C = 1 \Rightarrow\ P(C = 1, B = 0|do(D)) = P(B = 0)\sum_a P(A = a)P(C = 1|B = 0, A = a)$$
$$= 0.38$$

2.1.4. Using your answer to the previous problem, find $P(e|do(D)) = \sum_q P(Q = q, e|do(D))$

$$P(e|D) = P(B = 0|do(D)) = P(B = 0) = \sum_c P(C = c, B = 0)$$
$$= 0.12 + 0.38 = 0.5$$

2.1.5. Using your answers to the previous problems, find $P(Q|e, do(D)) = \frac{P(Q,e|do(D))}{P(e|do(D))}$

$$P(C = 0|B = 0, do(D)) = P(C = 0|B = 0) = \frac{P(C = 0, B = 0)}{P(B = 0)} = \frac{0.12}{0.5} = 0.24$$
$$P(C = 1|B = 0, do(D)) = P(C = 1|B = 0) = \frac{P(C = 1, B = 0)}{P(B = 0)} = \frac{0.38}{0.5} = 0.76$$

2.1.2. Using your result for $P(S|do(A), e)$ from the previous problem, find $EU(a|e)\ \forall\ a$.

$$EU(d|B = 0) = \sum_c P(c|B = 0)U(c, d)$$
$$EU(D = 0|B = 0) = P(C = 0|B = 0)U(C = 0, D = 0) + P(C = 1|B = 0)U(C = 1, D = 0)$$
$$= 0.24 * 100 + 0.76 * 40 = 54.4$$
$$EU(D = 1|B = 0) = P(C = 0|B = 0)U(C = 0, D = 1) + P(C = 1|B = 0)U(C = 1, D = 1)$$
$$= 0.24 * 20 + 0.76 * 80 = 65.6$$

2.1.3. Using your EU calculations above, find $MEU(B = 0)$ and $d^* = argmax_d EU(d|B = 0)$.

$$MEU(B = 0) = \max_d EU(d|B = 0) = 65.6$$
$$d^* = 1$$

Problem 3 – Value of Perfect Information

Using the network and your answer from the previous problem, we're going to compute the Value of Perfect Information (VPI) of knowing the state of variable A when $B = 0$ is given. Let's do so step-by-step:

3.1. Find $MEU(A = 0, B = 0)$ (hint: $P(S|do(A), e)$ is an immediately answerable query here).

$$MEU(A = 0, B = 0) = \max_d EU(d|A = 0, B = 0) = \max_d \sum_c P(c|A = 0, B = 0)U(c, d)$$

$$EU(D = 0|A = 0, B = 0) = \sum_c P(c|A = 0, B = 0)U(c, D = 0)$$

$$= P(C = 0|A = 0, B = 0)U(C = 0, D = 0) + P(C = 1|A = 0, B = 0)U(C = 1, D = 0)$$
$$= 0.8 * 100 + 0.2 * 40 = 88$$

$$EU(D = 1|A = 0, B = 0) = \sum_c P(c|A = 0, B = 0)U(c, D = 1)$$

$$= P(C = 0|A = 0, B = 0)U(C = 0, D = 1) + P(C = 1|A = 0, B = 0)U(C = 1, D = 1)$$
$$= 0.8 * 20 + 0.2 * 80 = 32$$

$$\therefore MEU(A = 0, B = 0) = 88$$

3.2. Using your answer to 3.1 and knowledge that $MEU(A = 1, B = 0) = 74$ (freebee!) find $MEU(A, B = 0)$.

We have to weight the $MEU(A = a, B = 0)$ by the likelihood of $A = a|B = 0$:

$$MEU(A, B = 0) = \sum_a P(a|B = 0)MEU(a, B = 0) = \sum_a P(a)MEU(a, B = 0)$$

$$= P(A = 0)MEU(A = 0, B = 0) + P(A = 1)MEU(A = 1, B = 0)$$
$$= 0.2 * 88 + 0.8 * 74 = 76.8$$

3.3. Compute the $VPI(A|B = 0)$.

$$VPI(A|B = 0) = MEU(A, B = 0) - MEU(B = 0) = 76.8 - 65.6 = 11.2$$

3.4. If the utility scores represent dollar amounts, what would be a fair price for A when $B = 0$?

The fair price is given by the VPI! That's its definition. So, the fair cost for info on A is: \$11.20. If we bought it for less, it would be a good deal, and for more, a ripoff.

CMSI 3300 – Classwork 5

SOLUTION

Instructions:
This worksheet will not only provide you with practice problems whose concepts will help you understand HW3 and your final exam, and think about the types of applications in which some of our Markov models can be deployed in practice.

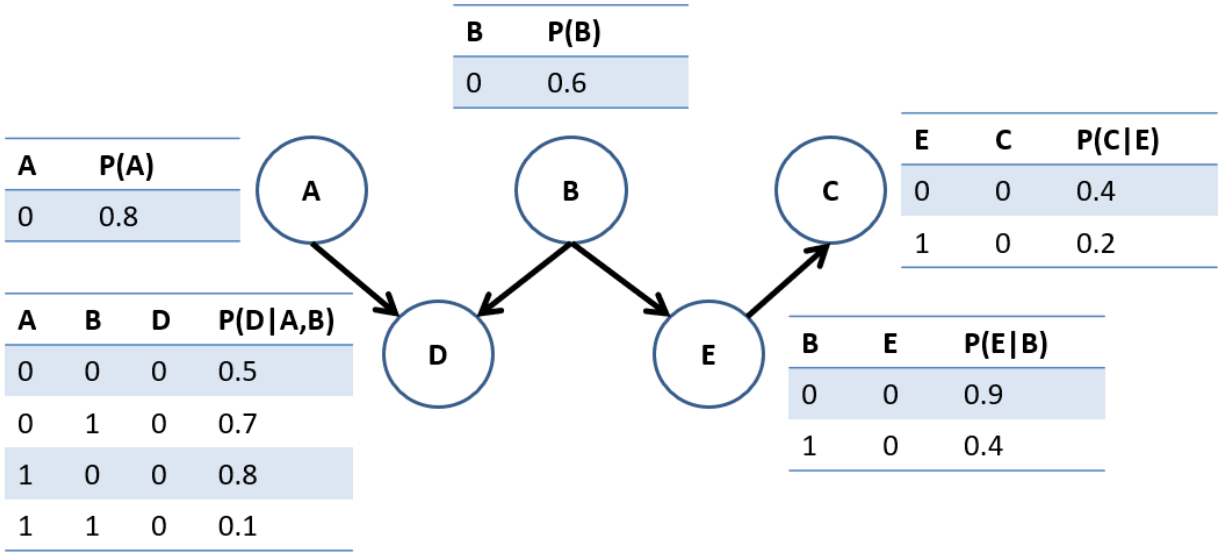
- Provide answers to each of the following questions and write your responses in the blanks. If you are expected to show your work in arriving at a particular solution, space will be provided for you.
- Place the names of your group members below:

Group Members:

1. _____
2. _____
3. _____

Problem 1 – Bayesian Network Approximate Inference

Consider the following Bayesian Network, and use it to answer the questions that follow:



B	P(B)
0	0.6

A	P(A)
0	0.8

E	C	P(C|E)
0	0	0.4
1	0	0.2

A	B	D	P(D|A,B)
0	0	0	0.5
0	1	0	0.7
1	0	0	0.8
1	1	0	0.1

B	E	P(E|B)
0	0	0.9
1	0	0.4

1.1. Provide a *possible* order in which variables would be sampled by *Prior Sampling:*

[A, B, D, E, C]
(among other possibilities, as long as parents are always sampled before their children)

Suppose now that Prior Sampling is used to generate the following set of 10 samples.

Sample	A	B	C	D	E
0	0	1	1	0	0
1	0	0	1	1	0
2	1	0	0	1	1
3	0	1	1	1	1
4	1	1	1	0	0
5	0	0	0	1	1
6	0	1	1	1	0
7	1	0	0	1	0
8	0	0	1	1	0
9	0	0	1	1	0

1.2. Using these samples, find $\hat{P}(C = 1|D = 1)$

Find the proportion of C = 1 samples among the total of those for which D = 1:

$$\hat{P}(C = 1|D = 1) = \frac{count(C = 1, D = 1)}{count(D = 1)} = \frac{5}{8}$$

1.3. Using the same model, let's trace the generation of Gibbs Sampling in order to estimate $\hat{P}(B|C=1,D=1,E=0)$. Suppose for the current sample we are sampling variables in the order of: $[A, B, \dots]$ and begin with the following random state. For each variable sampled thereafter, assume (for this exercise) that the most likely value of the distribution sampled-from is the one actually picked.

A	B	C	D	E
0	1	1	1	0

Sampling A (Provided as an Example)

1.3.1. From what distribution will A be sampled? Show the likelihoods of sampling each value.

Distribution Sampled: $P(A|B=1,C=1,D=1,E=0)$

$$P(A=0|B=1,C=1,D=1,E=0) = \frac{P(A=0)P(D=1|A=0,B=1)}{\sum_a P(A=a)P(D=1|A=a,B=1)}$$
$$= \frac{0.8*0.3}{0.8*0.3+0.2*0.9} \approx 0.57$$

1.3.2. Assuming our random number generator selects the value for A that is most likely, what will be the result of the above sample, and thus, the next state? (Fill in the value for A below):

A	B	C	D	E
0	1	1	1	0

Sampling B (Your Turn)

1.3.3. From what distribution will B be sampled? Show the likelihoods of sampling each value.

Distribution Sampled: $P(B|A=0,C=1,D=1,E=0)$

$$P(B=0|A=0,C=1,D=1,E=0)$$
$$= \frac{P(B=0)P(D=1|A=0,B=0)P(E=0|B=0)}{\sum_b P(B=b)P(D=1|A=0,B=b)P(E=0|B=b)}$$
$$= \frac{0.6*0.5*0.9}{0.6*0.5*0.9+0.4*0.3*0.4}$$
$$\approx 0.85$$

1.3.4. Assuming our random number generator selects the value for B that is most likely, what will be the result of the above sample, and thus, the next state? (Fill in the value for B below):

A	B	C	D	E
0	0	1	1	0

Problem 2 – Hidden Markov Models

In bioinformatics, analyses of nucleotide sequences in DNA processing is a central, but noisy, process; determining the identity of each nucleotide (from the 4 possible: ACGT) is not only imperfect, but many nucleotides in the sequence are "non-coding," meaning they do not encode any protein sequences (and are, for many applications, ignorable). It turns out the likelihood of producing certain nucleotides (viz., GC) increases during "coding" $(X = 1)$ sections, while the likelihoods of these two to express *decrease* during non-coding $(X = 0)$. From lab data, we have parameterized the following HMM describing this process:

Initial

X	$P(X_0)$
0	0.50

Transitions

X_t	X_{t+1}	$P(X_{t+1}\|X_t)$
0	0	0.60
1	0	0.50

Emissions

E_t	$P(E_t\|X_t = 0)$	$P(E_t\|X_t = 1)$
A	0.30	0.20
C	0.20	0.30
G	0.20	0.30
T	0.30	0.20

2.1. Suppose we observe a (small) sequence of nucleotides: GGC. Treating our time step t as position within the sequence, use filtering to compute $B(X_3)$, our belief of the current state after observing the entire sequence, starting with the following as given:

	$t = 0$ *Init*	$t = 1$ $E_1 = G$	$t = 2$ $E_2 = G$	$t = 3$ $E_3 = C$
$B(X_t = 0)$	0.5	0.4	0.44	0.443
$B(X_t = 1)$	0.5	0.6	0.56	0.557

Forward Algorithm (Combining both Updates for Time and Observation):

$$B(X_3 = 0) = \alpha\, P(E_3 = C|X_3 = 0) \sum_{x_2} P(X_3 = 0|X_2 = x_2)B(x_2)$$

$$= \alpha\, P(E_3 = C|X_3 = 0)[P(X_3 = 0|X_2 = 0)B(X_2 = 0) + P(X_3 = 0|X_2 = 1)B(X_2 = 1)]$$

$$= \alpha\, 0.2 * [0.6 * 0.44 + 0.5 * 0.56] = \alpha\, 0.1088$$

$$B(X_3 = 1) = \alpha\, P(E_3 = C|X_3 = 1)[P(X_3 = 1|X_2 = 0)B(X_2 = 0) + P(X_3 = 1|X_2 = 1)B(X_2 = 1)]$$

$$= \alpha\, 0.3 * [0.4 * 0.44 + 0.5 * 0.56] = \alpha\, 0.1368$$

$$\alpha = \frac{1}{0.1088 + 0.1368} = \frac{1}{0.2456}$$

$$B(X_3 = 0) = \frac{0.1088}{0.2456} \approx 0.443; \therefore B(X_3 = 1) = 1 - B(X_3 = 0) \approx 0.557$$

In this setting, since we really care about the *most likely state* at any given part of the sequence, it's often useful to seek an **explanation** of what led us to witness the evidence at any given time. However, to compute this *most likely explanation*, we need to amend our Forward Algorithm slightly; instead of summing over all *possible* previous states in the transitions, we focus on only the most likely, labeled $M(X_t)$, in what is known as the **Viterbi Algorithm**.

Forward Algorithm	*Viterbi Algorithm*
$B(X_{t+1}) = \alpha\, P(E_{t+1}\|X_{t+1}) \sum_{x_t} P(X_{t+1}\|x_t) B(x_t)$	$M(X_{t+1}) = \alpha\, P(E_{t+1}\|X_{t+1}) \max_{x_t} P(X_{t+1}\|x_t) M(x_t)$

2.2. Suppose we observe the sequence $GGCA$ and are using the Viterbi Algorithm to find the most likely explanation of the state at each point in its generation. Using the following partially-completed computations, find $M(X_4)$: (hint: it *is* as simple as it seems, that's why it's nice).

	$\boldsymbol{t=0}$ ***Init***	$\boldsymbol{t=1}$ $\boldsymbol{E_1=G}$	$\boldsymbol{t=2}$ $\boldsymbol{E_2=G}$	$\boldsymbol{t=3}$ $\boldsymbol{E_3=C}$	$\boldsymbol{t=4}$ $\boldsymbol{E_4=A}$
$M(X_t=0)$	0.5	0.4	0.4	0.4	0.6
$M(X_t=1)$	0.5	0.6	0.6	0.6	0.4

$$M(X_4=0) = \alpha\, P(E_4=A|X_4=0) \max_{x_3} P(X_4=0|X_3=x_3)\, M(X_3=x_3)$$
$$= \alpha\, 0.3 \max(0.6*0.4, 0.5*0.6)$$
$$= \alpha\, 0.09$$

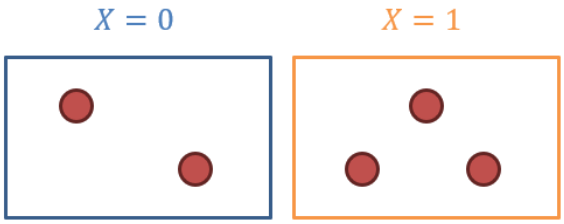
$$M(X_4=1) = \alpha\, P(E_4=A|X_4=1) \max_{x_3} P(X_4=1|X_3=x_3)\, M(X_3=x_3)$$
$$= \alpha\, 0.2 \max(0.4*0.4, 0.5*0.6)$$
$$= \alpha\, 0.06$$

$$\therefore \alpha = \frac{1}{0.15}$$
$$\therefore M(X_4=0) = 0.6;\ \ M(X_4=1) = 0.4$$

2.3. Examining the completed table above, determine the *explanation* (i.e., the most likely value of the state) for the observed sequence in all steps $t = 1 \rightarrow 4$:

	$\boldsymbol{t=1}$ $\boldsymbol{E_1=G}$	$\boldsymbol{t=2}$ $\boldsymbol{E_2=G}$	$\boldsymbol{t=3}$ $\boldsymbol{E_3=C}$	$\boldsymbol{t=4}$ $\boldsymbol{E_3=A}$
X_t^*	$X=1$	$X=1$	$X=1$	$X=0$

Consider instead that we decided to use HMM Approximate Inference with a Particle Filter (doesn't make a whole lot of sense because the state is so small here but meh, practice!) with $N = 5$ particles (again, very small). Consider that our approximate belief state $\hat{B}(X_t)$ is represented by the current allocation of particles as drawn below:

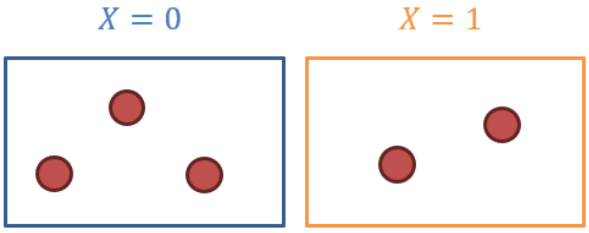


2.4. Suppose we now observe $E_{t+1} = T$ as the next nucleotide in the sequence, and we want to update our approximate belief state.

2.4.1. ***Elapse Time:*** What's the chance that each particle in the $X = 0$ state remain in this state?

Just check the proper transition distribution: $P(X_{t+1} = 0 | X_t = 0) = 0.6$

Suppose after elapsing time, we are left with the following configuration of particles:



2.4.2. ***Observe Evidence:*** Labeling the particles in the above diagram from left-to-right as $S^{(0)}, S^{(1)}$, …, how are each weighted after observing $E_{t+1} = T$?

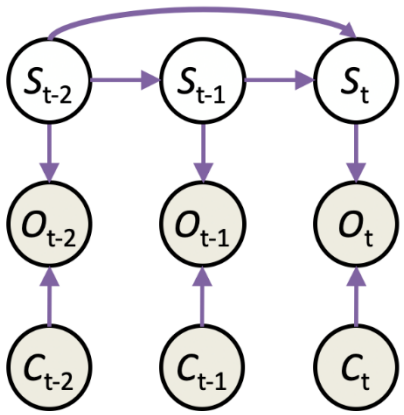
	$S^{(0)}$	$S^{(1)}$	$S^{(2)}$	$S^{(3)}$	$S^{(4)}$
$w(S^{(i)})$	0.3	0.3	0.3	0.2	0.2

2.4.3. ***Resample:*** What's the likelihood that each particle is resampled at $X = 0$ and $X = 1$?

$P\left(S_{t+1}^{(i)} = 0 \mid E_{t+1} = T, S_t^{(i)} = s_t^{(i)}\right) = \frac{0.9}{1.3}$	$P\left(S_{t+1}^{(i)} = 1 \mid E_{t+1} = T, S_t^{(i)} = s_t^{(i)}\right) = \frac{0.4}{1.3}$

In Hideo Kojima's reboot of Pacman, it is the year 2077, and the hunt between Pacman and ghosts has become more technologically dystopian. In this psychological thriller, Pacman is trying to hunt ghosts in an infinite hallway with array-like index positions labeled as in the diagram below. As it's the future, Pacman's got some new gadgets, and wants to estimate the ghost's **actual position, S_t**, using some additional flashy sensors (the entire budget of the game was used to animate them). From the sensors, Pacman can find, at each time step, **a noisy reading of the ghost's location, O_t**. However, the ghost too has gained new cloaking technology, such that if **it cloaks itself at any given time step C_t**, it adds extra noise to the sensor readings.

Noting that, at each time step, the sensors must calibrate to not only the previous step, but the one before that as well, Pacman generates the following modified HMM:



State S_t: Corridor Index

...	4	5	6	7	8	...

Transition Model

D_1	D_2	$P(S_t \mid S_{t-1}, S_{t-2})$
0	0	0.6
0	1	0.2
0	2	0.0
1	0	0.3
1	1	0.3
1	2	0.5

$D_1 = |S_t - S_{t-1}|$
$D_2 = |S_t - S_{t-2}|$

Emission Model

C_t	D_t	$P(O_t \mid S_t, C_t)$
0	0	0.4
0	1	0.2
0	2	0.1
1	0	0.6
1	1	0.2
1	2	0.0

$D_t = |O_t - S_t|$

Given that our belief $B(S_t)$ relies on two previous states, were we to use *Particle Filtering* in this problem, we would need each particle to hold two states. For example, the particle denoting a vote that the ghost was in index 6 at time $t = 6$ and then at index 7 at $t = 7$ is: $(S_6 = 6, S_7 = 7)$

2.5 Suppose we have a particle $(S_6 = 6, S_7 = 7)$; what is the probability that, when accounting for passage of time, the particle transitions to $(S_7 = 7, S_8 = 8)$? (Show how you arrive at your answer using the model above).

$$(S_6 = 6, S_7 = 7) \rightarrow (S_7 = 7, S_8 = 8) \Rightarrow D_1 = 1, D_2 = 2$$
$$P(S_8 = 8 \mid S_6 = 6, S_7 = 7) = 0.5$$

2.6 Independent from the previous step, suppose we observe the following:

$$C_6 = 1, \quad O_6 = 6, \quad C_7 = 0, \quad O_7 = 8$$

When we update the likelihood of particles for these observations, what would be the weight w associated with a particle $(S_6 = 7, S_7 = 8)$? [Hint: examine what we are given in the emission model, and observe that we would have to incorporate essentially 2 updates rolled into 1 given that our particles consist of 2 states; this translates to a simple product of each step's emission].

$P(O_6 = 6, O_7 = 8 \mid C_6 = 1, S_6 = 7, C_7 = 0, S_7 = 8) \Rightarrow D_6 = 1, D_7 = 0$ $= P(O_6 = 6 \mid C_6 = 1, S_6 = 7) * P(O_7 = 8 \mid C_7 = 0, S_7 = 8) = 0.2 * 0.4 = 0.08$
$w\big((S_6 = 7, S_7 = 8)\big) = 0.08$

Ignoring the previous tracking of a single particle, suppose we now have the following set of weighted particles, having already updated for time and observations, but have yet to resample:

Particle p	**$w(p)$**
$(\boldsymbol{S_7 = 5, S_8 = 6})$	0.10
$(\boldsymbol{S_7 = 7, S_8 = 6})$	0.25
$(\boldsymbol{S_7 = 7, S_8 = 7})$	0.30

2.7 *Prior to resampling*, suppose we renormalize the weights associated with each state the "old-fashioned" way like in exact inference, what would be Pacman's belief state at $t = 8$? Complete the table below to indicate your answer, showing work for each cell [Hint: just look at S_8 in the particles above and the weights attached to each]:

$Position, S_8$	$\hat{B}(S_8)$
$S_8 = 5$	0
$S_8 = 6$	$\frac{0.1 + 0.25}{0.1 + 0.25 + 0.3} = \frac{0.35}{0.65} = \frac{7}{13}$
$S_8 = 7$	$\frac{0.3}{0.1 + 0.25 + 0.3} = \frac{0.3}{0.65} = \frac{6}{13}$
$S_8 = 8$	0

CMSI 3300 – Classwork 6

SOLUTION

Instructions:
This worksheet will not only provide you with practice problems for your upcoming exam, but will add to your toolset as initiate data scientists taking various machine learning problems from start to finish. Specific notes:

- Provide answers to each of the following questions and write your responses in the blanks. If you are expected to show your work in arriving at a particular solution, space will be provided for you.
- Place the names of your group members below:

Group Members:

1. _____
2. _____
3. _____

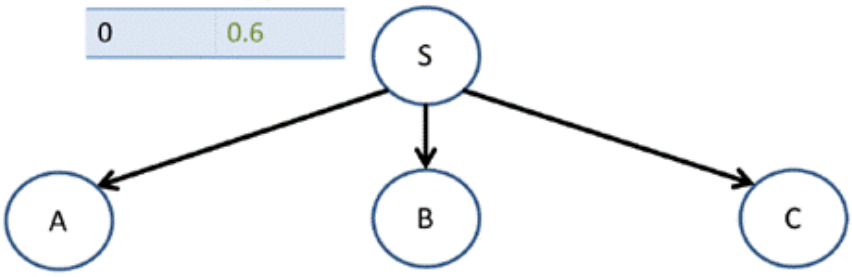
Problem 1 – Naïve Bayes Classifiers

Forney Industries' FornGene group has continued work developing a classifier to identify cases of Schistoforneymiosis [S] based on indicators of three genetic markers [A, B, C]. In this section, we will attempt to craft a Naïve Bayes Classifier that explains the data collected by the FornGene group, and use that to answer classification queries related to the disease. Note how this problem is different than the "Bag of Words" example from class in that each feature's CPT has its own parameters.

S	A	B	C	# of datum
0	0	0	0	48
0	0	0	1	144
0	0	1	0	12
0	0	1	1	36
0	1	0	0	72
0	1	0	1	216
0	1	1	0	18
0	1	1	1	54
1	0	0	0	21
1	0	0	1	189
1	0	1	0	7
1	0	1	1	63
1	1	0	0	9
1	1	0	1	81
1	1	1	0	3
1	1	1	1	27

1.1: In the table above, we track each combination of A, B, C, and S witnessed from the group's study. From this data, construct the *Maximum Likelihood* CPTs over each feature below. *Place your final answers directly in the tables below.*

S	P(S)
0	0.6



S	A	P(A | S)
0	0	0.40
1	0	0.70

S	B	P(B | S)
0	0	0.80
1	0	0.75

S	C	P(C | S)
0	0	0.25
1	0	0.10

1.2. Using the data from the previous section, suppose we instead *smooth* our estimates for the feature likelihoods using Laplacian Smoothing with $k = 10$. Compute $P_{LAP,10}(C = 0|S = 1)$

$$P_{LAP,10}(C = 0|S = 1) = \frac{c(C = 0, S = 1) + k}{c(S = 1) + k|X|}$$
$$= \frac{40 + 10}{400 + 10 * 2}$$
$$\approx 0.12$$

[!] Note: |X| = 2 here because the variable to which we are artificially adding samples is C, which is binary. Since we're adding $k = 10$ samples to each of $C = 0,1$, that's why we add 20 extra to the total in the denominator!

1.3: Using the NBC you learned in Part 1.1, determine (showing your work) the most likely class for each of the following data points.

1. $\{A = 0, B = 1, C = 0\}$

$$P(S = 0|A = 0, B = 1, C = 0) \propto P(S = 0)P(A = 0|S = 0)P(B = 1|S = 0)P(C = 0|S = 0)$$
$$= 0.6 * 0.4 * 0.2 * 0.25$$
$$= 0.012$$

$$P(S = 1|A = 0, B = 1, C = 0) \propto P(S = 1)P(A = 0|S = 1)P(B = 1|S = 1)P(C = 0|S = 1)$$
$$= 0.4 * 0.7 * 0.25 * 0.1$$
$$= 0.007 \text{ (in Die Another Bayes)}$$

$$P(S = 0|A = 0, B = 1, C = 0) > P(S = 1|A = 0, B = 1, C = 0)$$
$$\therefore Choose\ S = 0$$

2. $\{A = 1, C = 1\}$ [Hint: Remember that NBCs are still Bayesian Networks, so B as a missing feature shouldn't be a problem – make sure your answer derives why]

$$P(S|A, C) \propto \sum_b P(S)P(A|S)P(b|S)P(C|S) = P(S)P(A|S)P(C|S)\sum_b P(b|S) = P(S)P(A|S)P(C|S)$$
$$P(S = 0|A = 1, C = 1) \propto P(S = 0)P(A = 1|S = 0)P(C = 1|S = 0)$$
$$= 0.6 * 0.6 * 0.75$$
$$= 0.270$$

$$P(S = 1|A = 1, C = 1) \propto P(S = 1)P(A = 1|S = 1)P(C = 1|S = 1)$$
$$= 0.4 * 0.3 * 0.9$$
$$= 0.108$$

$$P(S = 0|A = 1, C = 1) > P(S = 1|A = 1, C = 1)$$
$$\therefore Choose\ S = 0$$

Problem 2 – Linear Perceptrons

Returning to our email classification supervised learning task, consider that we have a trinary class variable $Y \in \{0,1,2\} = \{Spam, Ham, Important\}$. Moreover, we've decided on a simple feature extractor f that takes an email x as input and returns a vector of features indexed as:

1. $f_0(x) =$ *# of ALL CAPS words*
2. $f_1(x) =$ *# of occurrences of word "free"*
3. $f_2(x) =$ *whether or not email is in known contacts (0 = not in contacts, 1 = in contacts)*

2.1. What feature vector would the above feature extractor return for the following email?

$Email\ (x)$	$f_0(x)$	$f_1(x)$	$f_2(x)$
From: Ray.Toal@lmu.edu *Message:* Hi all, There is free pizza in the Keck Lab, COME GET IT!	3	1	1

Consider that the above is in a training set with label $y = 2$ (this is a very important email). If, during learning, our class weight vectors are as follows…

$w_0 = \langle 2, 2, -3 \rangle$ $w_1 = \langle -1, 1, 2 \rangle$ $w_2 = \langle 2, -3, 1 \rangle$

2.2. First, determine which class y our perceptron would currently assign this email.

$$w_0 \cdot f(x) = 2 * 3 + 2 * 1 + (-3) * 1 = 5$$
$$w_1 \cdot f(x) = -1 * 3 + 1 * 1 + 2 * 1 = 0$$
$$w_2 \cdot f(x) = 2 * 3 + (-3) * 1 + 1 * 1 = 4$$

$$y = argmax_y\ w_y \cdot f(x) = 0$$

2.3. Did our Perceptron make a mistake? If so, calculate the updated weights that would amount from its misclassification. If not, draw BlindBot on vacation (or draw him in either case if you want, I'm a classwork, not a cop).

Nope, bad Perceptron! It said $y = 0$ when the correct was $y = 2$. As such, we diminish the $y = 0$ weights and enhance the $y = 2$ weights proportionate to the feature vector that caused the error:

$$w_0 = w_0 - f(x) = \langle 2,2,-3 \rangle - \langle 3,1,1 \rangle = \langle -1,1,-4 \rangle$$
$$w_2 = w_2 + f(x) = \langle 2,-3,1 \rangle + \langle 3,1,1 \rangle = \langle 5,-2,2 \rangle$$

Problem 3 – Logistic Regression

Suppose now we pivot to a multiclass logistic regression model with $Y \in \{0,1,2\}$ and the following class-specific weights.

$$w_0 = \langle 2, 2, -3 \rangle \qquad w_1 = \langle -1, 1, 2 \rangle \qquad w_2 = \langle 2, -3, 1 \rangle$$

3.1. Compute the likelihoods of each class $P\left(\widehat{Y^{(i)}} \middle| x^{(i)}; w\right)$ that a Logistic Regression classifier would give for the sample features $f\left(x^{(i)}\right) = \langle 1,1,1 \rangle$. Show your work / steps in the box below.

$$z_0 = w_0 \cdot f(x) = \langle 2,2,-3 \rangle \cdot \langle 1,1,1 \rangle = 2 + 2 - 3 = 1$$
$$z_1 = w_1 \cdot f(x) = \langle -1,1,2 \rangle \cdot \langle 1,1,1 \rangle = -1 + 1 + 2 = 2$$
$$z_2 = w_2 \cdot f(x) = \langle 2,-3,1 \rangle \cdot \langle 1,1,1 \rangle = 2 - 3 + 1 = 0$$

$$e^{z_0} = e^1 \approx 2.72$$
$$e^{z_1} = e^2 \approx 7.39$$
$$e^{z_2} = e^0 = 1.00$$

$$P\left(\widehat{Y^{(i)}} = 0 \middle| x^{(i)}; w\right) = \frac{e^{z_0}}{e^{z_0} + e^{z_1} + e^{z_2}} = \frac{2.72}{11.11} \approx 0.25$$
$$P\left(\widehat{Y^{(i)}} = 1 \middle| x^{(i)}; w\right) = \frac{e^{z_1}}{e^{z_0} + e^{z_1} + e^{z_2}} = \frac{7.39}{11.11} \approx 0.67$$
$$P\left(\widehat{Y^{(i)}} = 2 \middle| x^{(i)}; w\right) = \frac{e^{z_2}}{e^{z_0} + e^{z_1} + e^{z_2}} = \frac{1.00}{11.11} \approx 0.09$$

$P\left(\widehat{Y^{(i)}} = 0 \middle\| x^{(i)}; w\right) =$	$P\left(\widehat{Y^{(i)}} = 1 \middle\| x^{(i)}; w\right) =$	$P\left(\widehat{Y^{(i)}} = 2 \middle\| x^{(i)}; w\right) =$
0.25	0.67	0.09

3.2. Suppose our sample $x^{(i)}$ was meant to have label $y^{(i)} = 0$. Determine the new weights that would be associated with each class in a single step of stochastic gradient descent with learning rate $\eta = 0.1$ and cross entropy loss. Show your work in the box below.

$$\frac{\delta L_{CE}}{\delta w_0} = -\left[1 - P\left(\widehat{Y^{(i)}} = 0 \middle| x^{(i)}; w\right)\right] * f(x)$$
$$= -[1 - 0.25] * \langle 1,1,1 \rangle$$
$$= \langle -0.75, -0.75, -0.75 \rangle$$
$$\frac{\delta L_{CE}}{\delta w_1} = -\left[0 - P\left(\widehat{Y^{(i)}} = 1 \middle| x^{(i)}; w\right)\right] * f(x)$$
$$= -[0 - 0.67] * \langle 1,1,1 \rangle$$
$$= \langle 0.67, 0.67, 0.67 \rangle$$
$$\frac{\delta L_{CE}}{\delta w_2} = -\left[0 - P\left(\widehat{Y^{(i)}} = 2 \middle| x^{(i)}; w\right)\right] * f(x)$$
$$= -[0 - 0.09] * \langle 1,1,1 \rangle$$
$$= \langle 0.09, 0.09, 0.09 \rangle$$

$$w_0 = w_0 - \eta \frac{\delta L_{CE}}{\delta w_0} = \langle 2,2,-3 \rangle - 0.1\langle -0.75, -0.75, -0.75 \rangle = \langle 2.075, 2.075, -2.925 \rangle$$
$$w_1 = w_1 - \eta \frac{\delta L_{CE}}{\delta w_1} = \langle -1,1,2 \rangle - 0.1\langle 0.67, 0.67, 0.67 \rangle = \langle -1.067, 0.933, 1.933 \rangle$$
$$w_2 = w_2 - \eta \frac{\delta L_{CE}}{\delta w_2} = \langle 2,-3,1 \rangle - 0.1\langle 0.09, 0.09, 0.09 \rangle = \langle 1.991, -3.009, 0.991 \rangle$$

$w_0 =$ $\langle 2.075, 2.075, -2.925 \rangle$	$w_1 =$ $\langle -1.067, 0.933, 1.933 \rangle$	$w_2 =$ $\langle 1.991, -3.009, 0.991 \rangle$